

MATM20 Mathematical Modelling

Project Report

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Introduction

In this project, we will study a model called Predator-Prey to model the population development between a rabbit population and fox population.

A rabbit population and a fox population live in contact. The foxes feed on the rabbits, which will limit the growth of the rabbit population and facilitate the growth of the fox population. Throughout this report, we will refer concepts taught from the main textbook of the course. [1] We are given the following information: the rabbit population R is modelled using an intrinsic growth rate of $r_R = 2$ and a maximum sustainable population K_R . This is a measure of the strength of resource limitations. The growth rate function g_R for rabbits is the following:

$$g_R = r_R R \left(1 - \frac{1}{K_R} R \right).$$

The growth of the rabbits is further limited by rabbits being killed by foxes. This is modelled as:

$$h_{FR} = r_R \alpha R F.$$

The α is estimated to be $\alpha = 0.01$. Here F denotes the fox population. The rabbit population is reduced when F becomes large. In this model the fox population is limited by death from natural causes. The loss rate of the foxes is estimated as:

$$l_F = b_F F.$$

With a coefficient $b_F = 1/5$. The fox population F can only grow by feeding on rabbits. The growth is affected by the number of rabbits and the number of foxes. The larger both populations the faster the fox population will grow. The growth rate function for foxes is:

$$g_F = \frac{1}{B} R F.$$

It is estimated $B = 1000$.

The rabbits correspond to the prey and the foxes to the predators. If there are no preys, the fox population would go extinct. If both are present the cause

of death of the predator would be natural death, while for the prey, it would be being eaten by a fox.

Question 1

In this question we will first determine which values of K_R the two species can coexist.

Let R and F denote the populations of rabbits and foxes, respectively. We will use the five-step method.

Step 1:

Before we can ask a question in mathematical terms we go through the problem and make a list of variables and assumptions.

Variables:

R = rabbits population
 F = fox population
 g_R = growth rate for rabbits
 g_F = growth rate for foxes
 h_{FR} = loss of rabbits
 l_F = loss of foxes

Assumptions:

$$\begin{aligned} g_R &= r_R R \left(1 - \frac{1}{K_r} R\right) \\ g_F &= \frac{1}{B} R F \\ h_{FR} &= r_R \alpha R F \\ l_F &= b_F F \end{aligned}$$

Here, r_R, B and b_F are positive reals.

Objective:

To find values of K_R for which the two species can coexist.

Step 2:

In this step, we will select the modeling approach. We will model this problem as a dynamical model in steady state.

Step 3:

This step is to formulate the model. Let R and F denote our two state variables, defined on the state space:

$$\{(R, F) : R \geq 0, F \geq 0\}.$$

The steady-state equations are:

$$\begin{aligned} r_R R \left(1 - \frac{1}{K_R} R \right) - r_R \alpha R F &= 0 \\ -b_F F + \frac{1}{B} R F &= 0 \end{aligned}$$

With the values given, we can write them as:

$$\begin{aligned} 2R \left(\left(1 - \frac{1}{K_R} R \right) - 0.01F \right) &= 0 \\ -\frac{1}{5}F + \frac{1}{1000}RF &= 0 \end{aligned}$$

We are interested in solutions of this system of equations that lie in the state space. These solutions represent the equilibrium points of our dynamical model.

Step 4:

This step is to solve the model. We find three solutions of the above system of equations:

$$O = (0, 0), \quad P = (K_R, 0), \quad Q = \left(200, 100 \left(1 - \frac{1}{K_R} 200 \right) \right)$$

These are the equilibrium points.

O: This is equivalent to say that both the populations are 0. After drawing the graph, and analysed the velocity vectors we came to the conclusion that the point is unstable, as the velocity arrows go away from it.

P: At this point, the population F is equal to zero, and the population R is stable. To determine if the point is stable or not we should investigate further, looking for eigenvalues.

Q: In the last solution we encounter the term $\left(1 - \frac{1}{K_R} 200 \right)$. We have three cases:

$$1. \quad \frac{1}{K_R} 200 = 1 \implies K_R = 200$$

If $K_R = 200$ then the point will be $(200, 0)$. In this case, foxes and rabbits cannot coexist.

$$2. \quad \frac{1}{K_R} 200 > 1 \implies K_R < 200$$

If $K_R < 200$, we will have a negative point, that will drive us outside the

state space, so we will not consider that.

$$3. \frac{1}{K_R} 200 < 1 \implies K_R > 200$$

If $K_R > 200$, we have a positive point, so it is an equilibrium point. By Fig.2, looking at the velocity vectors, we can guess that is a stable spiral. So, only in this case of $K_R > 200$, Q is a stable point

Only for the case 3, $K_R > 200$, we can say that we have three equilibrium points, otherwise we have just two equilibrium points. From that we can conclude that the two population can coexist for $K_R > Bb_F = 200$, otherwise the fox population will die because of lack of rabbits, or the rabbit population will go extinct because there are too many foxes.

Fig.1 shows the equilibria for the problem when $K_R > 200$ while Fig.2 shows its vector field. The velocity vectors show the direction in which the trajectories move.

As shown in Fig.2, We can divide the graph into four parts [3]:

- In the first part, the rabbit population will grow ($\frac{dR}{dt} > 0$), but the fox population will decrease ($\frac{dF}{dt} < 0$), so all velocity vectors go to the right and down.
- In the second part, the rabbit population grow ($\frac{dR}{dt} > 0$) and so the fox one ($\frac{dF}{dt} > 0$), the velocity vectors will go up and right.
- In the third part, the rabbit population will decrease ($\frac{dR}{dt} < 0$) while the fox one will grow ($\frac{dF}{dt} > 0$), the vector will go to the left and up.
- And in the fourth part, both the populations will decrease ($\frac{dR}{dt} < 0$, $\frac{dF}{dt} < 0$), the velocity vectors are going down and left.

The velocity vectors point to the the equilibrium point that is stable. Fig.3 and Fig.4 show the graph in case $K_R = 200$ and $K_R < 200$ respectively.

Step 5:

This step is to to summarize the results of our model analysis in non mathematical terms and answer the question posed originally in Step 1. Q is an equilibrium point that is consistently stable in the case $K_R > 200$. Thus, that would conclude that for values $K_R > 200$, the rabbit population would be big enough for both population to coexist. Notice that for $K_R = 200$ we are not able derive a conclusion for P, but for $K_R < 200$ the point P is stable.

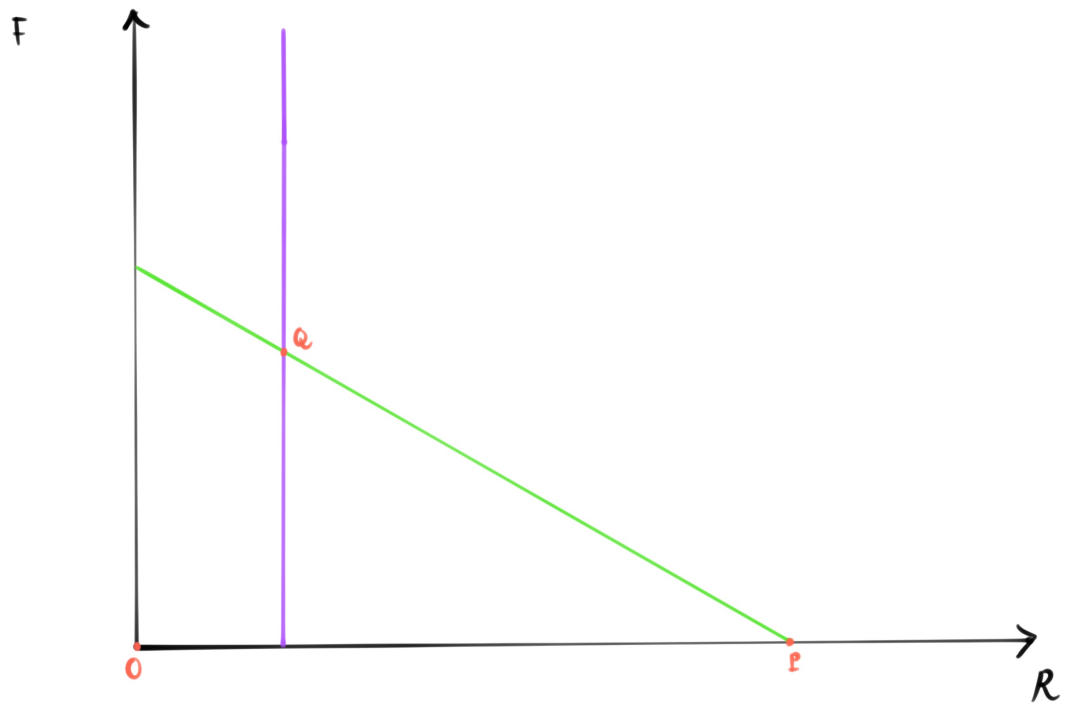


Fig. 1. Equilibrium points

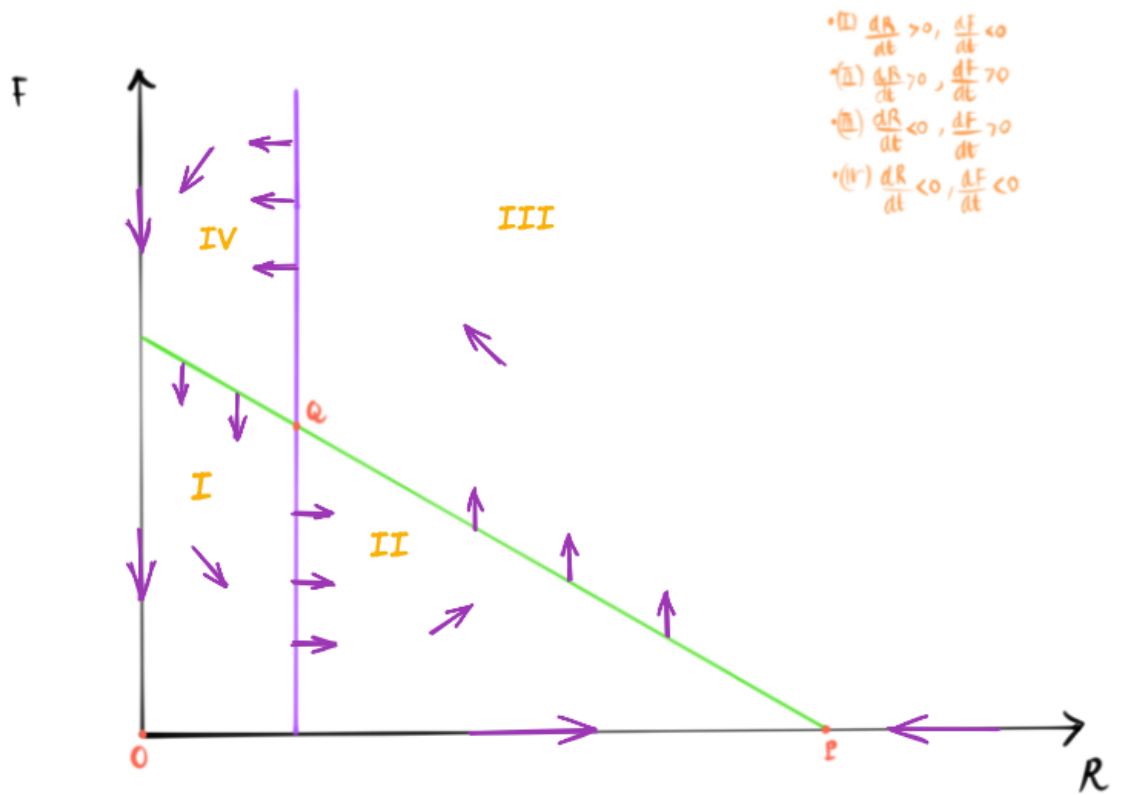


Fig. 2. With velocity vectors

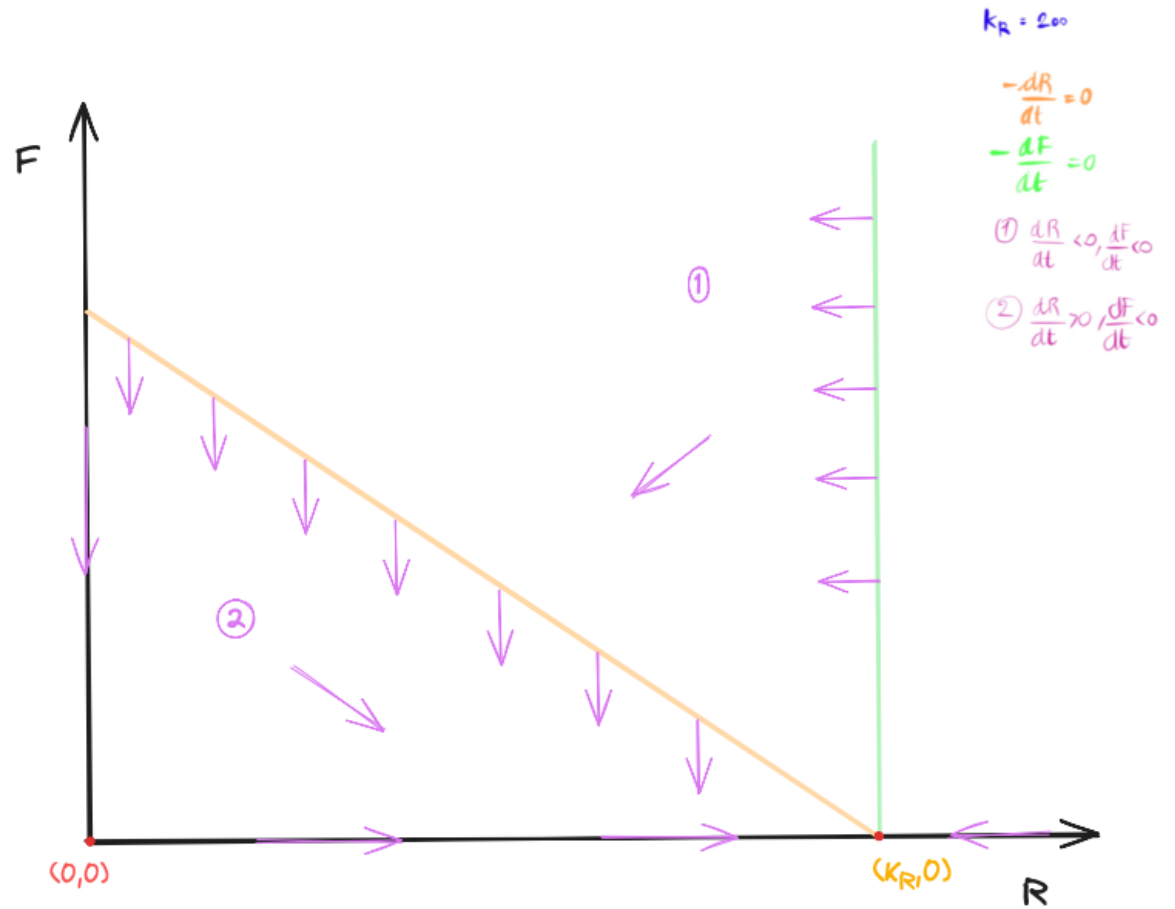


Fig. 3. $K_R = 200$

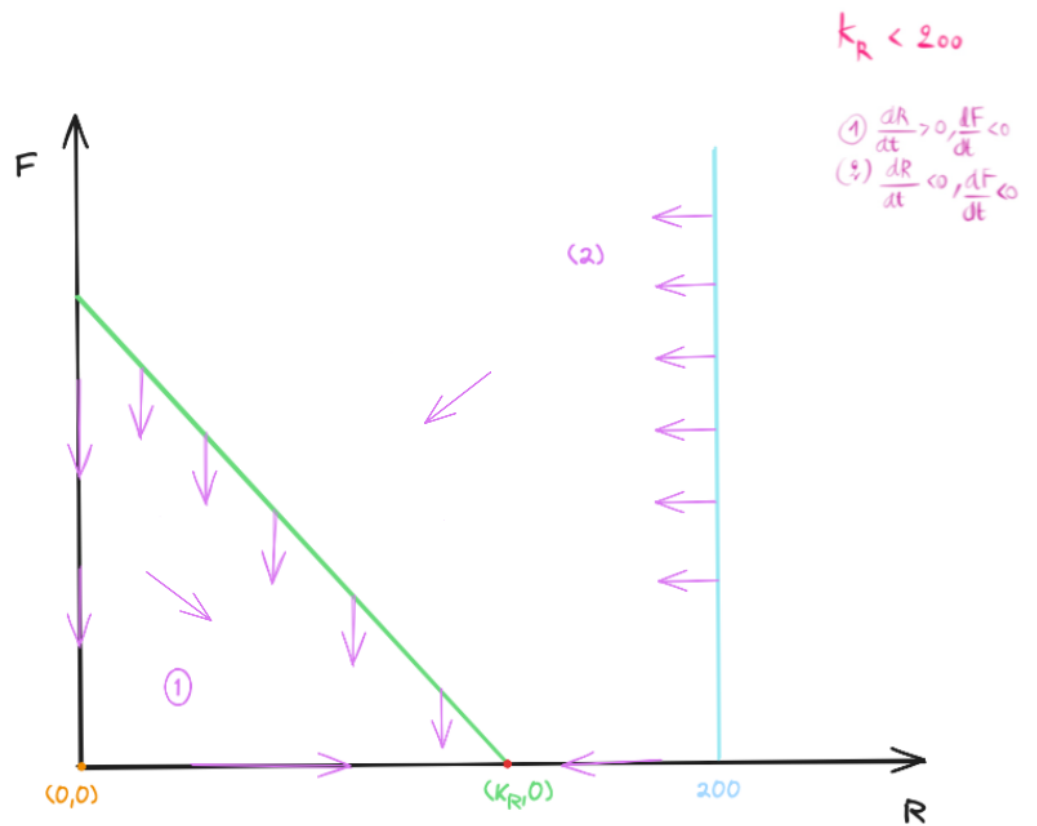


Fig. 4. $K_R < 200$

Question 2

In this question we will determine the stability of the equilibria using the eigenvalue method and investigate for $K_R \in \{1400, 3000\}$.

We are given a dynamical system of the form $x' = F(x)$ as the steady-state equations in **Question 1**.

A theorem states that an equilibrium point is stable if the matrix

$$\begin{pmatrix} \frac{\partial f_1}{\partial R} & \frac{\partial f_1}{\partial F} \\ \frac{\partial f_2}{\partial R} & \frac{\partial f_2}{\partial F} \end{pmatrix}$$

has eigenvalues with all negatives real parts. If any eigenvalue has a positive real part, then the equilibrium is unstable.[1] [2]

We have the following system:

$$\begin{aligned} 2R \left(\left(1 - \frac{1}{K_R} R \right) - 0.01F \right) &= 0 = f_1 \\ -\frac{1}{5}F + \frac{1}{1000}RF &= 0 = f_2 \end{aligned}$$

The partial derivatives are:

$$\begin{aligned} \frac{\partial f_1}{\partial R} &= 2 - 4\frac{1}{K_R}R - 0.02F \\ \frac{\partial f_1}{\partial F} &= -0.02R \\ \frac{\partial f_2}{\partial R} &= \frac{1}{1000}F \\ \frac{\partial f_2}{\partial F} &= -\frac{1}{5} + \frac{1}{1000}R \end{aligned}$$

Evaluating the partial derivatives at the equilibrium points. We obtain the following matrices:

At $O = (0, 0)$ we have:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{5} \end{pmatrix};$$

At $P = (K_R, 0)$ we have:

$$B = \begin{pmatrix} -2 & -0.02K_R \\ 0 & -\frac{1}{5} + \frac{1}{1000}K_R \end{pmatrix};$$

At $Q = (200, 100(1 - \frac{1}{K_R}200))$ we have:

$$C = \begin{pmatrix} 2 - 4\frac{200}{K_R} - 0.02 * 100(1 - \frac{1}{K_R}200) & -0.02 * 200 \\ \frac{1}{1000} * 100(1 - \frac{1}{K_R}200) & -\frac{1}{5} + \frac{1}{1000}200 \end{pmatrix}.$$

Simplifying this,

$$C = \begin{pmatrix} -\frac{400}{K_R} & -4 \\ 0.1 - \frac{20}{K_R} & 0 \end{pmatrix}.$$

In order to find the eigenvalues we compute the characteristic equation. For this question, we can find the characteristic equation and then the eigenvalues, or we can use the Python code in Appendix A to find the eigenvalues.

For matrix A, the eigenvalues can be seen straight away from the diagonal. They are $\lambda_1 = 2$, $\lambda_2 = -\frac{1}{5}$.

If $K_R = 1400$,

the eigenvalues of B are $\lambda_1 = -2$, $\lambda_2 = 1.2$;
the eigenvalues of C are

$$\begin{aligned} \lambda_1 &= -0.143 - 0.576i \\ \lambda_2 &= -0.143 + 0.567i. \end{aligned}$$

If $K_R = 3000$,

the eigenvalues of B are $\lambda_1 = -2$, $\lambda_2 = 2.8$;
the eigenvalues of C are

$$\begin{aligned} \lambda_1 &= -0.0667 - 0.607i \\ \lambda_2 &= -0.0667 + 0.607i. \end{aligned}$$

This drives us to the conclusion that the only stable points are the ones obtained for $K_R = 1400, 3000$ using the matrix C. While the eigenvalues of B, in both cases, are one positive and one negative, this drives us to the conclusion that P is unstable. A similar consideration holds for the matrix A, leading us to the conclusion that O is unstable.

In case of $K_R = 3000$ the stable point is: (200, 93.3), in case of $K_R = 1400$, the stable point is: (200, 85.7).

Fig.5 and Fig.6 show the phase portraits of the cases $K_R = 1400$ and $K_R = 3000$ respectively.

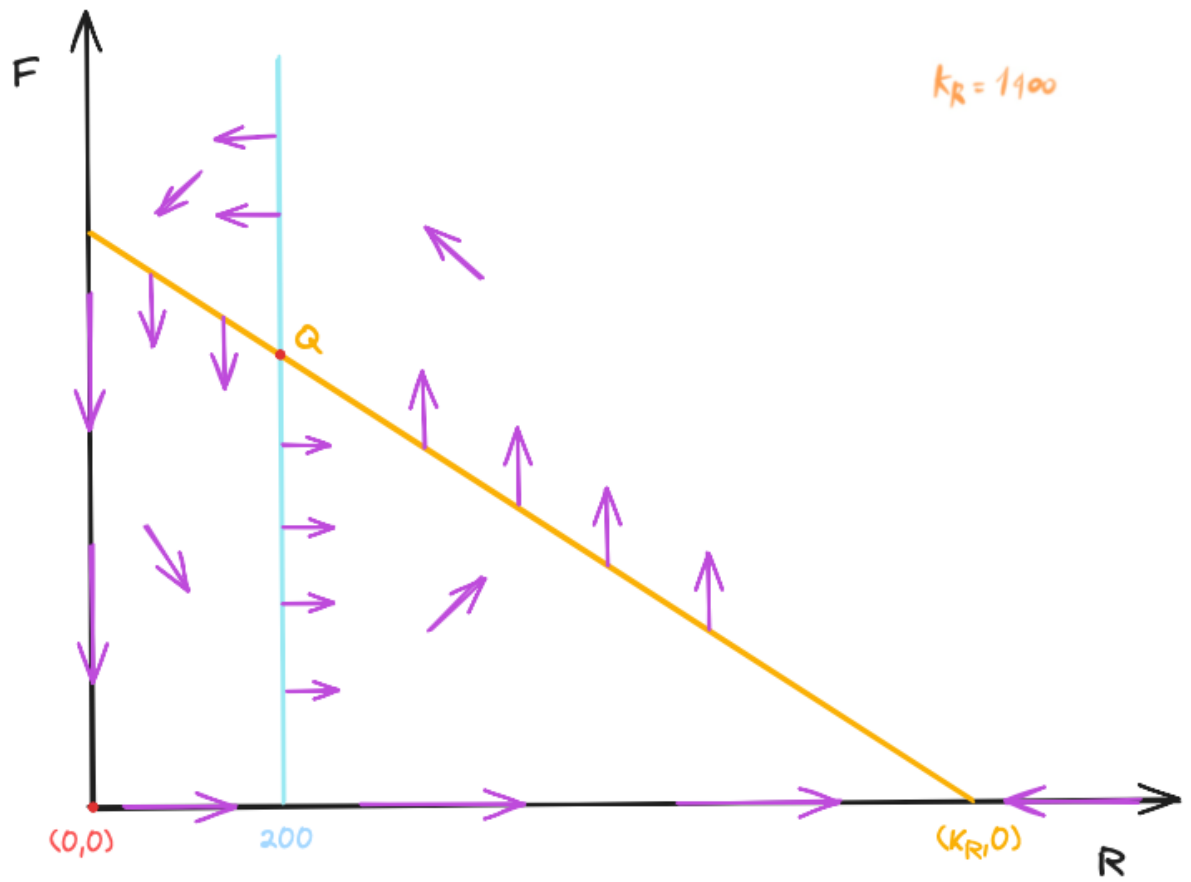


Fig. 5. $K_R = 1400$

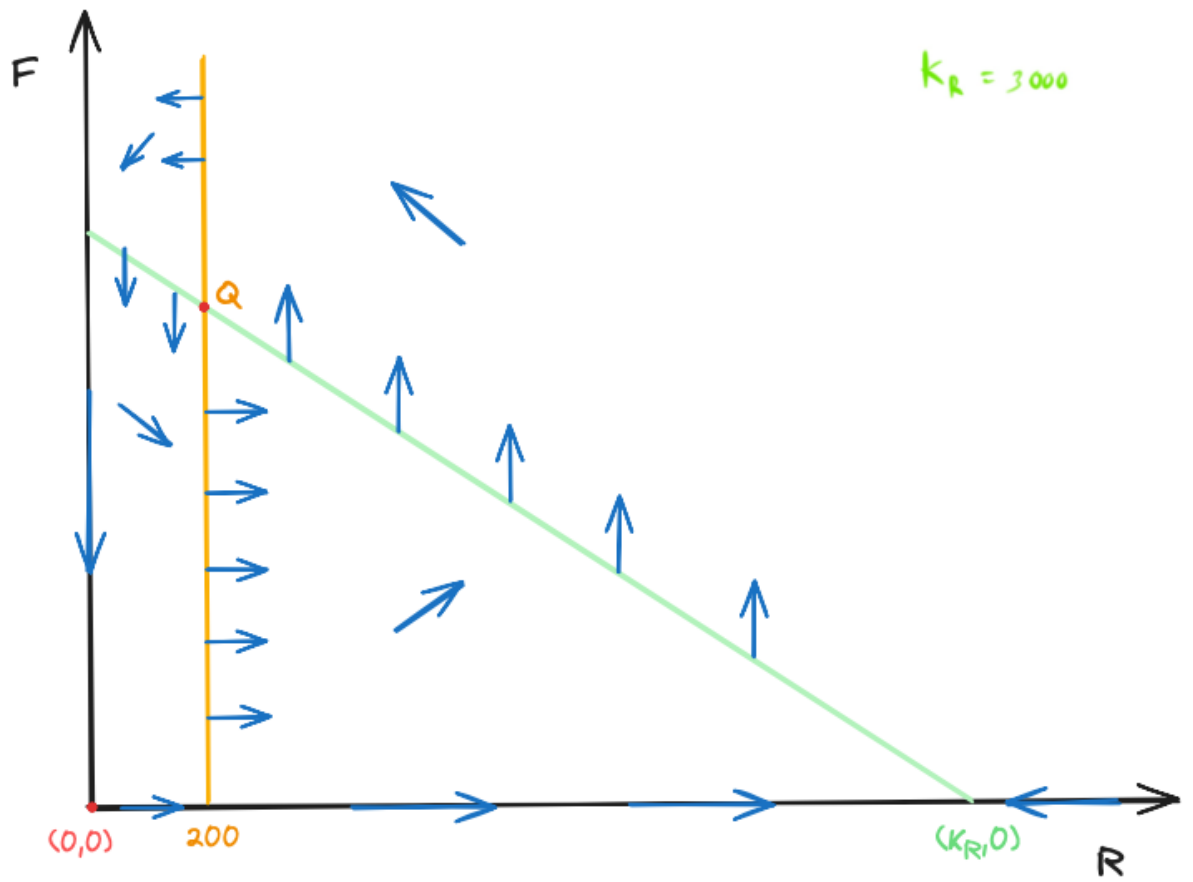


Fig. 6. $K_R = 3000$

Question 3

In this question we will perform a stability analysis for K_R and find out how far K_R can be increased or decreased until the solution changes qualitatively.

To perform a stability analysis for K_R we will investigate how the solutions change while changing K_R . Consider the following system of equations:

$$\begin{aligned} 2R \left(\left(1 - \frac{1}{K_R} R \right) - 0.01F \right) &= 0 = f_1 \\ -\frac{1}{5}F + \frac{1}{1000}RF &= 0 = f_2 \end{aligned}$$

The partial derivatives are:

$$\begin{aligned} \frac{\partial f_1}{\partial R} &= 2 - 4\frac{1}{K_R}R - 0.02F \\ \frac{\partial f_1}{\partial F} &= -0.02R \\ \frac{\partial f_2}{\partial R} &= \frac{1}{1000}F \\ \frac{\partial f_2}{\partial F} &= -\frac{1}{5} + \frac{1}{1000}R \end{aligned}$$

We will evaluate the partial derivatives at the equilibrium points and then obtain the following matrices:

At $O = (0, 0)$ we have:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{5} \end{pmatrix};$$

At $P = (K_R, 0)$ we have:

$$B = \begin{pmatrix} -2 & -0.02K_R \\ 0 & -\frac{1}{5} + \frac{1}{1000}K_R \end{pmatrix};$$

At $Q = (200, 100(1 - \frac{1}{K_R}200))$ we have:

$$C = \begin{pmatrix} 2 - 4\frac{200}{K_R} - 0.02 * 100(1 - \frac{1}{K_R}200) & -0.02 * 200 \\ \frac{1}{1000} * 100(1 - \frac{1}{K_R}200) & -\frac{1}{5} + \frac{1}{1000}200 \end{pmatrix}.$$

The last one can be simplified as:

$$C = \begin{pmatrix} -\frac{400}{K_R} & -4 \\ 0.1 - \frac{20}{K_R} & 0 \end{pmatrix}.$$

For matrix A, the eigenvalues can be seen straight away from the diagonal, they are $\lambda_1 = 2$, $\lambda_2 = -\frac{1}{5}$. O is unstable.

As seen in **Question 1**, we notice that the solution changes qualitatively if $K_R \leq 200$. Indeed in this case we will have just two equilibrium points, while for $K_R > 200$ we will have three equilibrium points. Just in the case of $K_R \leq 200$ the two populations cannot coexist.

Consider first the case of $K_R = 200$.

In this case the eigenvalues for matrix B are $\lambda_1 = -2$, $\lambda_2 = 0$. One of the eigenvalue is zero, so it's a degenerate case. The investigation for the point $(200, 0)$ is then inconclusive. [4].

In the case $K_R < 200$, for instance for $K_R = 20$, for matrix B we will have the following eigenvalues $\lambda_1 = -2$, $\lambda_2 = -0.18$. So the point P is stable. But in this case just the rabbit populations survives, and the fox population will die.

Lastly for $K_R > 200$, we can take $K_R = 600$, for the matrix C we will have the following eigenvalues $\lambda_1 = -0.333 - 0.394i$, $\lambda_2 = -0.333 + 0.394i$, as the real part of both eigenvalues is negative, we can conclude in this case that Q is a stable point. For matrix B, the eigenvalues are: $\lambda_1 = -2$, $\lambda_2 = 0.4$, the point $(600, 0)$ is unstable.

Question 4

In this question we will find out what will happen if $K_R = 900$ or in the limit $K_R \rightarrow \infty$.

If $K_R = 900$, then:

- It doesn't affect $O = (0, 0)$.
- $Q = (200, 77.7)$. Matrix C has eigenvalues $\lambda_1 = -0.222 - 0.512i$, $\lambda_2 = -0.222 + 0.512i$. So Q is a stable point.
- $P = (900, 0)$. Matrix B has eigenvalues $\lambda_1 = -2$, $\lambda_2 = 0.7$. So P is an unstable point.

Fig.7 show the phase portrait for $K_R = 900$.

If K_R tends to infinity then the value of $\frac{1}{K_R}$ tends to zero. With this new information, we can rewrite the system of equations from **Question 1** as follows:

$$\begin{aligned} 2R((1) - 0.01F) &= 0 = f_1 \\ -\frac{1}{5}F + \frac{1}{1000}RF &= 0 = f_2 \end{aligned}$$

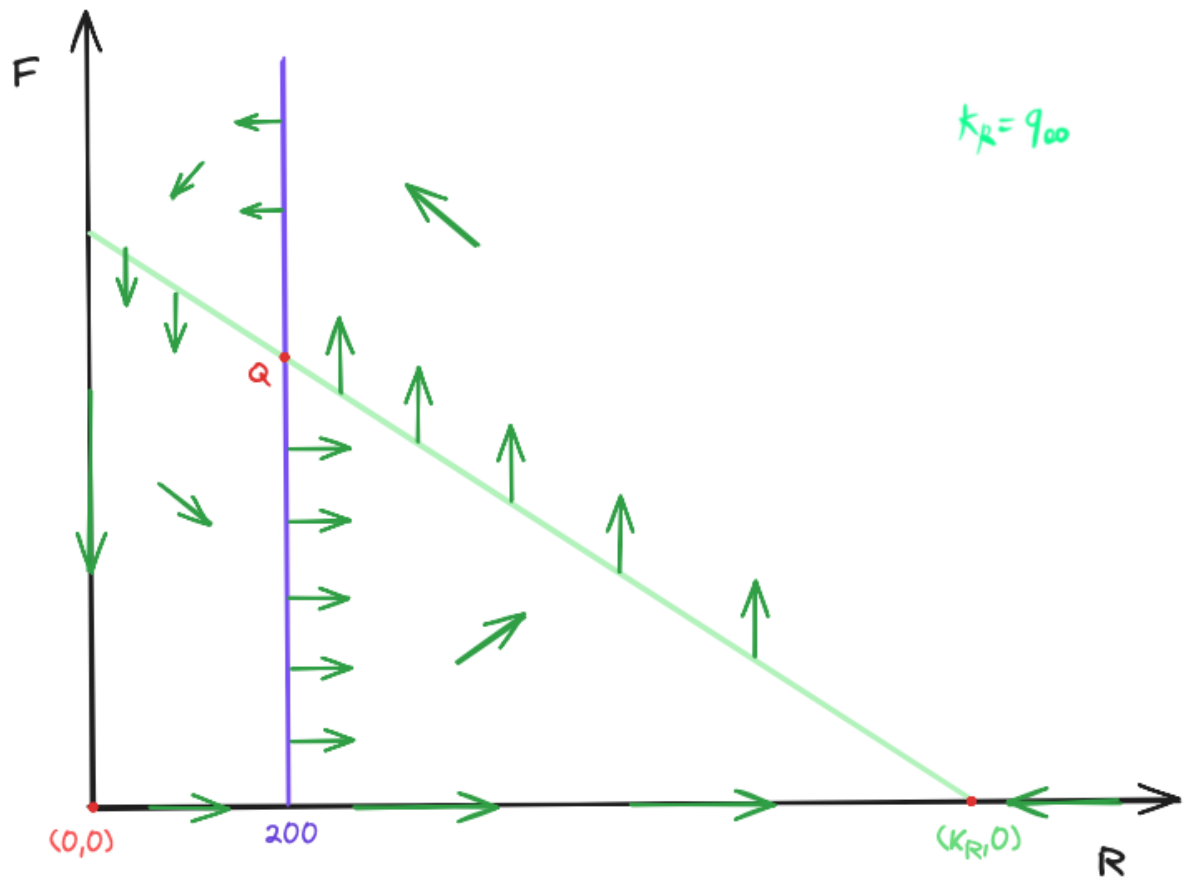


Fig. 7. $K_R = 900$

The equilibrium points are: $(0,0)$, $(200,100)$.

The partial derivatives are:

$$\begin{aligned}\frac{\partial f_1}{\partial R} &= 2 - 0.02F \\ \frac{\partial f_1}{\partial F} &= -0.02R \\ \frac{\partial f_2}{\partial R} &= \frac{1}{1000}F \\ \frac{\partial f_2}{\partial F} &= -\frac{1}{5} + \frac{1}{1000}R\end{aligned}$$

This leads us to the following matrix:

$$\begin{pmatrix} 2 - 0.02F & -0.02R \\ \frac{1}{1000}F & -\frac{1}{5} + \frac{1}{1000}R \end{pmatrix}.$$

At $O = (0,0)$, the matrix A will have the same eigenvalues as before. O is unstable.

At $Q = (200,100)$ we will have:

$$C = \begin{pmatrix} 0 & -4 \\ 0.1 & 0 \end{pmatrix}$$

So the eigenvalues in this case are given by:

$$(C - \lambda I) = \begin{pmatrix} -\lambda & -4 \\ 0.1 & -\lambda \end{pmatrix}$$

We can thus find the characteristic equation:

$$\det(C - \lambda I) = \lambda^2 + 0.4$$

Hence, the eigenvalues are $\lambda_1 = \sqrt{0.4}i$, $\lambda_2 = -\sqrt{0.4}i$.

We have just pure imaginary values with real parts equal to 0.

As we increase the values of K_R to infinity, it can be noticed that the real part of eigenvalues tends to zero. If the real part of the complex eigenvalues are 0, then that would imply that the velocity vectors form the shape of ellipses around the equilibrium point. This equilibrium point would be stable. An arbitrarily small perturbation of the point could change it into a stable or unstable spiral. [2] Thus, if K_R goes to infinity, the real part of the eigenvalues obtained from C matrix are zero which would lead to a phase portrait of ellipses with the equilibrium point as its centre.

A Appendix

Python Code:

Note for reader: This code is useful to calculate eigenvalues.

```

1 import sympy as sp
2 x1, x2, k_x1 = sp.symbols('x1,x2,k_x1')
3 x1,x2,k_x1
4 f1=2*x1*(1-1/k_x1*x1)-2*0.01*x1*x2
5 f1
6 f2=-1/5*x2+1/1000*x1*x2
7 f2
8 sol = sp.solve([sp.simplify(f1/x1),sp.simplify(f2/x2)], [x1,x2
9                                     ])
10 sol
11 df1_dx1 = sp.diff(f1,x1)
12 df1_dx2 = sp.diff(f1,x2)
13 df2_dx1 = sp.diff(f2,x1)
14 df2_dx2 = sp.diff(f2,x2)
15 A=sp.Matrix([
16     [df1_dx1, df1_dx2],
17     [df2_dx1, df2_dx2]
18 ])
19 A
20 A_eq = A.subs([(x1,sol[x1]), (x2,sol[x2])])
21 A_eq
22 sp.simplify(A_eq)
23 Ev_dict = A_eq.eigenvals()
24 Ev_dict
25 Ev1= sp.simplify(list(Ev_dict)[0])
26 Ev1
27 Ev2= sp.simplify(list(Ev_dict)[1])
28 Ev2
29 ev1_num=Ev1.subs(k_x1, 3000)
30 ev1_num
31 ev2_num=Ev2.subs(k_x1, 3000)
32 ev2_num

```

References

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